

Judy Effendi

<https://judyeffendi.github.io/judyeffendi/>
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Professional Summary

Software Engineer with a B.S. in Computer Science and Biology. Experienced in backend development, data engineering, web development, and clinical settings. Passionate about applying bioinformatics, AI, and software tools to improve patient outcomes and support precision medicine research.

Work Experience

University of Michigan Medicine

Ann Arbor, MI

Clinical Research Coordinator

July 2025- Present

- Conduct clinical research in global pathology, digital pathology, artificial intelligence, and medical education under Dr. Rouba Ali-Fehmi.
- Analyze clinical and international program data and collaborate with faculty, biostatisticians, and global partners across 25+ countries.
- Contributed to 3+ peer-reviewed publications and abstracts through data analysis, manuscript preparation, and research dissemination.
- Coordinate GPALMC research initiatives and support the development of international pathology education and research programs.

NAAMA NextGen

Ann Arbor, MI

Web Developer and Research Lead

July 2025- Present

- Developed and maintained the NAAMA NextGen website using HTML, CSS, and Java, supporting a network of 1,000+ members.
- Moderated AI in Medicine panels, facilitating discussions at the intersection of engineering, artificial intelligence, and clinical medicine.
- Led the NAAMA NextGen Research Symposium, coordinating research presentations and programming.
- Chaired the Media and Communications Committee for the NAAMA International Conference, overseeing digital outreach, promotional strategy, and conference communications.

Target Corp.

Minneapolis, MN

Software Engineer

September 2023-May 2025

- Developed a Kotlin based module for Kafka to achieve seamless data mirroring from production to stage.
- Implemented robust redaction mechanisms to safeguard sensitive data from being propagated to output topics in a scrum agile environment.
- Performed unit testing of endpoints to validate functionality and security measures for authorization mechanisms and maintained and optimized PostgreSQL DB for efficient data storage and retrieval.

Bosch

East Lansing, MI

Software Engineer, Capstone Experience

January 2023- April 2023

- Developed a Web Interface for HIL (hardware-in-the-loop) Benches to centralize remote access for various teams across Bosch utilizing a Flask API for an instance of a HIL that sends HTTP requests.
- Created and implemented Python-based automated test suites to test vehicle control modules and collect the necessary logs to enable development teams to fix defects and prevent safety issues in production.

Target Corp.

Remote

Software Engineer Intern

June 2022-August 2022

- Leveraged the power of Spark/Scala to develop a cutting-edge automated data comparator tool for the Data Engineering Stores Planning team, streamlining data analysis and validation processes.
- Implemented an end-to-end real-life enterprise product utilized by Target-Tech

College of Engineering, MSU

East Lansing, MI

CSE Outreach Administrator

August 2020-April 2023

- Exemplified **leadership** as one of many **lead admins** of our esteemed local and international computer science outreach programs.
- Successfully introduced the dynamic fields of Python programming and Web Development to a diverse range of students from 6th-12th grade, with an emphasis on empowering underrepresented demographics.

Computer Science Department, College of Engineering, MSU

East Lansing, MI

Undergraduate CSE ULA

January 2022-April 2023

- Actively guided a class of 30 students in debugging, problem-solving exercises, labs, and projects.

- Effectively communicated complex technical concepts in a clear and concise manner, catering to diverse learning styles and enabling students to grasp programming principles efficiently.

College of Engineering, MSU

East Lansing, MI

Ambassador, Women in Engineering

January 2020-April 2023

- Led and administered K-12 outreach activities to recruit and encourage girls to pursue careers within the engineering field.
- Encouraged opportunities for academic, personal, and professional growth by organizing, leading, and planning a 6-week Leadership Series for 30+ undergraduate Women in Engineering

Relevant Skills

- Languages: Python, SQL, Spark/Scala, Kotlin, C++, ReactJS, HTML/CSS, Flask, REST APIs
- Data/Cloud: PostgreSQL, Grafana, Google Cloud, AWS, Docker, Kafka
- Bioinformatics-Related Tools: Jupyter Notebooks, Pandas, scikit-learn, matplotlib
- DevOps/Other: Git, Jira, JUnit, Vela
- Personal: Leadership, Adaptability

Education

Michigan State University

August 2019-April 2023

B.S., Computer Science and Engineering, Minor: Spanish

University of Michigan

Post-Baccalaureate Biology